

Reflective Journal

This project had challenged us to adapt a classification dataset (CIFAR-10) for object detection, a task that initially seemed daunting. The objective was to improve upon a baseline model using a lightweight architecture, all within the constraints of Google collabs tier. The dataset that was selected to go with (CIFAR-10) was primarily because of its readily available format and small size, crucial considering the computational limitations. However, this choice also presented unique hurdles .

Initially, I anticipated difficulties in adapting CIFAR-10, given its inherent design for image classification. The images are relatively small ex. (32x32), and importantly, lack bounding box annotations - the essential data for object detection training. To overcome this, I had to generate synthetic bounding boxes. I with the help of my partner were able to develop/formulate a simple algorithm that assumed a central object and generated bounding boxes based on estimated object size. This approach was a compromise and I acknowledged that along with the limitations in the accuracy of the generated boxes. The lack of true bounding box annotations almost certainly introduced a source of error in the final model accuracy.

I started with the SSD mobilenet V2 model, following the structure of the notebook provided. I found along with the group that the transfer learning aspects of the exercise were particularly useful as the pre-trained weights allowed me to leverage knowledge from a larger dataset and focus on fine-tuning to the specifics of my adapted CIFAR-10 dataset. Computationally intensive, taking several hours to train and optimize the hyperparameters.

The attempt to improve the model using data augmentation techniques (random flips, rotations, and brightness) proved successful. The augmented dataset greatly increased the robustness of the final model. However even with the augmentation, the limitations of the synthetic bounding boxes and small image size became apparent. The final mAP was noticeably lower than what would be expected on a naturally annotated object detection dataset of comparable size. I would say that i would attribute this lower performance to mainly due to the inaccurate location of the bounding boxes generated by the program.